

# Derivatives — group rounds

Lessons 1–12 · Quarter I · 40 min · 16 questions · 80 points · groups of 3–4

Four rounds, each timed and worth more than the last. Work on paper and hand in one sheet for the group. An answer with no method scores half.

GROUP

MEMBERS

SCORE

Round A · Easy5'2 pts each

One step each — speed round

- 1

Find the slope of the secant to  $y = x^2$  between  $x = 1$  and  $x = 3$ .
- 2

Evaluate  $\lim$  as  $x \rightarrow 2$  of  $3x^2 - 5$ .
- 3

Evaluate  $\lim$  as  $x \rightarrow 4$  of  $\frac{x^2 - 16}{x - 4}$ .
- 4

Differentiate  $y = x^8$ .
- 5

Differentiate  $y = 5x^3 - 2x$ .
- 6

Differentiate  $y = (3x + 1)^4$ .

Round B · Medium7'4 pts each

Two or three steps

- 7

Evaluate  $\lim$  as  $x \rightarrow 0$  of  $\frac{\sqrt{x+9} - 3}{x}$ .
- 8

Use first principles to find  $f'(x)$  for  $f(x) = x^2 - 5x$ .
- 9

Differentiate  $y = \frac{2x - 3}{x + 1}$ .
- 10

Differentiate  $y = \sqrt{x^2 + 7}$ .
- 11

Where does  $y = x^3 - 12x$  have a horizontal tangent?

Round C · Hard8'6 pts each

Two rules at once — be ready to explain

- 12

Differentiate  $y = x^3(2x - 5)^4$  and factorise the answer.
- 13

Evaluate  $\lim$  as  $x \rightarrow 3$  of  $\frac{x^2 - x - 6}{x^2 - 9}$ .
- 14

Find the equation of the tangent to  $y = \sqrt{4x + 9}$  at  $x = 4$ .

Round D · Very hard8'15 pts each

Both parts needed for full marks

- 15

A spherical balloon is inflated so that its **surface area** grows at  $24 \text{ cm}^2/\text{s}$ .  
(a) Find the rate the radius grows when  $r = 3 \text{ cm}$ . (b) Find the rate the volume grows at that moment.
- 16

(a) Use first principles to show that  $f(x) = \frac{1}{x}$  has  $f'(x) = -\frac{1}{x^2}$ .  
(b) Hence find the equation of the **normal** to  $y = \frac{1}{x}$  at  $x = 2$ .